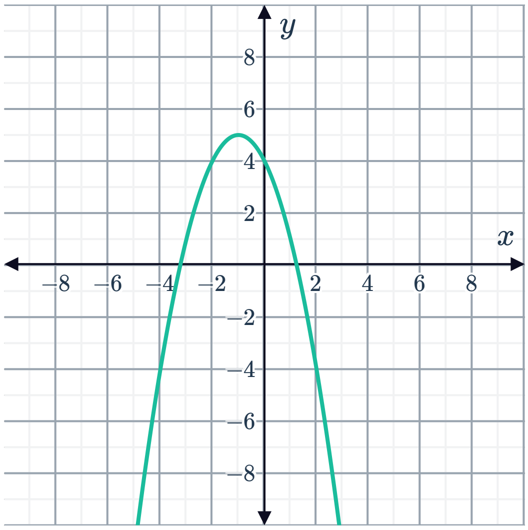
Quadratic equations and parabolas

1 For the following graphs of functions of the form $y = ax^2 + bx + c$:

- **i** State whether the vertex of the parabola is a maximum or minimum point.
- **ii** State whether the value of a is negative or positive.
- **iii** State the number of solutions to the equation $ax^2 + bx + c = 0$.

a



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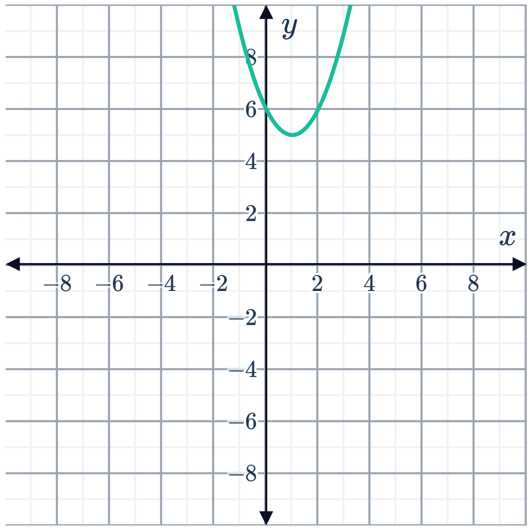

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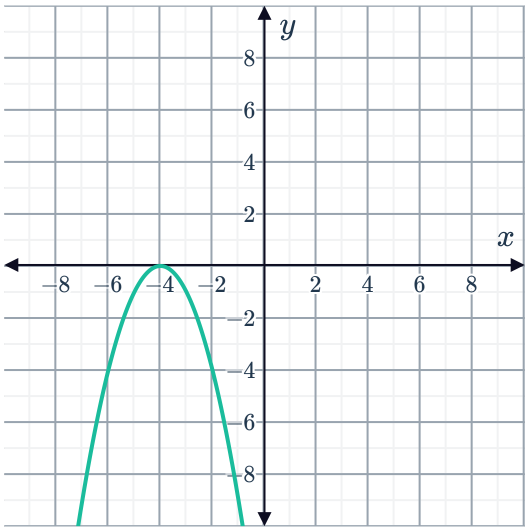

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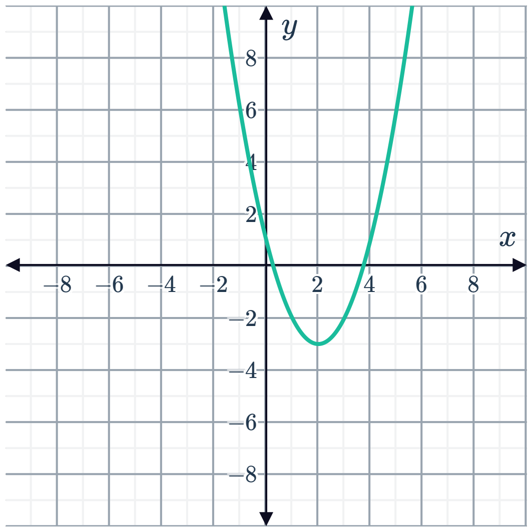


2 Below is the result after using the quadratic formula to solve an equation:

$$x = \frac{-(-10) \pm \sqrt{-1}}{8}$$

What can be concluded about the solutions of the equation?

3 By inspection, determine the number of real solutions for each of the following quadratic equations:

a $x^2 = 9$ **b** $(x-4)^2 = 0$ **c** $(x-6)^2 = -2$

The discriminant and equations

4 For the following quadratic equations:

i Find the value of the discriminant.

ii Hence state the number of real solutions for the equation.

a $x^2+6x+9=0$

b $4x^2-6x+7=0$

c $2x^2-2x=x-1$

d $x^2-4=0$

e $-x^2-8x-16=0$

f $-2x^2+5x-6=0$

g $-x^2+6x+1=0$

h $-x^2+25=0$

5 Consider the equation $x^2+22x+121=0$.

a Find the value of the discriminant.

b State whether the solutions to the equation are rational or irrational.

6 For the following equations:

i Find the value of the discriminant.

ii State the nature of the roots.

a $5x^2+4x+8=0$

b $4x^2+4x-6=0$

c $4x^2-4x+1=0$

d $x^2-16=0$

e $9x^2-42x=-49$

f $2x^2-9x=-2$

g $x^2=-6x-10$

h $\frac{1}{4}x^2-2x+\frac{1}{4}=0$

7 Consider the equation $x^2-8x-48=0$.

a Find the discriminant.

b Describe the nature of the roots.

c Find the solutions of the equation.

8 For a particular quadratic equation $b^2-4ac=0$, what can be said about the solutions of the quadratic equation?

9 The solutions of a quadratic equation are 9 and -9. What can be said about the value of b^2-4ac?

10 Find an expression for the discriminant of [redacted]llowing quadratic equations:

a $mx^2 + 3x - 2 = 0$

b $x^2 - 4nx - 2 = 0$

c $x^2 + 5x + p - 5 = 0$

d $-px^2 + 4x + p = 0$

e $mx^2 - 2mx + m - 3 = 0$

f $\dfrac{3n}{4}x^2 + \dfrac{1n}{2}x + 2n = 0$

11 Find the range of values of a for which the quadratic equation $ax^2 - bx + c = 0$, where $b = 6$ and $c = 10$, has two distinct real solutions.

12 Consider the equation in terms of x:

$$mx^2 + 2x - 1 = 0$$

Given that it has two unique solutions, determine the possible values of m.

13 Consider the equation in terms of x:

$$mx^2 - 3x - 5 = 0$$

a Given that the equation has two unique solutions, determine the possible values of m.

b State the value of m that must be eliminated from the range of solutions found in the previous part. Explain your answer.

14 Find the values of n for which $x^2 - 8nx + 1296 = 0$ has one solution.

15 Find the value of k for which $16x^2 + 8x + k = 0$ has equal roots.

16 Find the values of m for which $(m + 4)\,x^2 + 2mx + 2 = 0$ has a single solution.

17 For the following equations:

i Find the discriminant in terms of k.

ii Find the value of k so that the equation has equal solutions.

iii Find the value of k so that the equation has real solutions.

iv Find the value of k so that the equation has no real solutions.

v Find the value of k so that the equation has real and distinct solutions.

a $2x^2 + 8x + k = 0$

b $kx^2 + 20x + 2 = 0$

c $(k + 4)\,x^2 + 10x + 3 = 0$

18 Consider the equation in terms of x:

$$x^2 + 18x + k + 7 = 0$$

a Find the values of k for which the equation has no real solutions.

b State the smallest possible integer value of k.

The discriminant and parabolas

19 When graphing a particular parabola, Katrina used the quadratic formula and found that $b^2 - 4ac = -5$. How many x-intercepts does the parabola have?

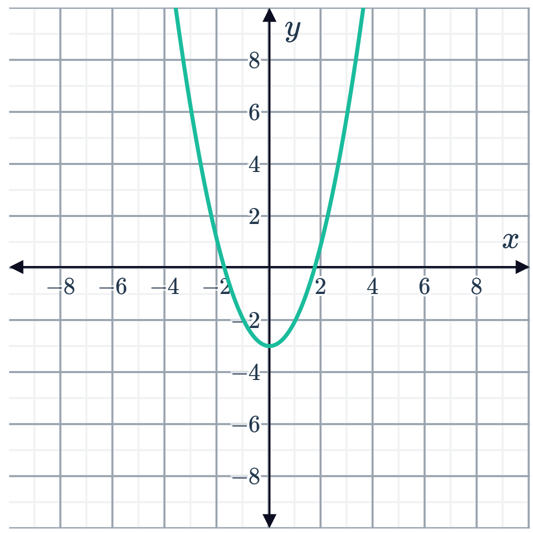
20 When graphing a particular parabola, Tony used the quadratic formula and found that $b^2 - 4ac = 0$. How many x-intercepts does the parabola have?

21 For each of the following graphs of a quadratic $f(x) = ax^2 + bx + c$, with discriminant $\Delta = b^2 - 4ac$:

i State whether $a > 0$ or $a < 0$.

ii State whether $\Delta > 0$, $\Delta < 0$ or $\Delta = 0$.

a



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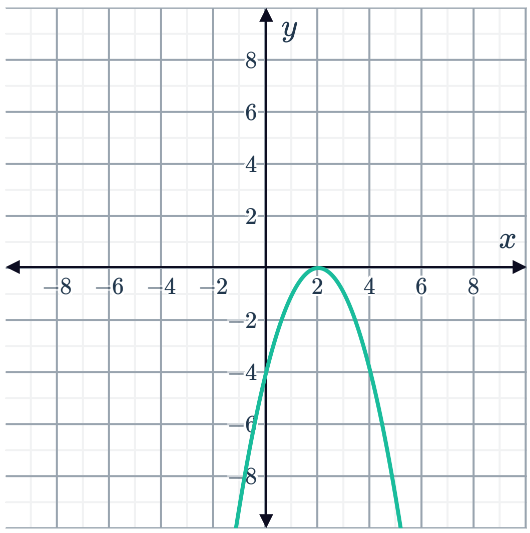

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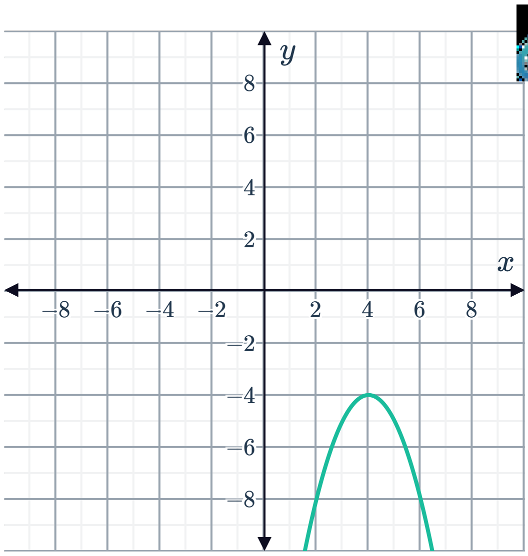
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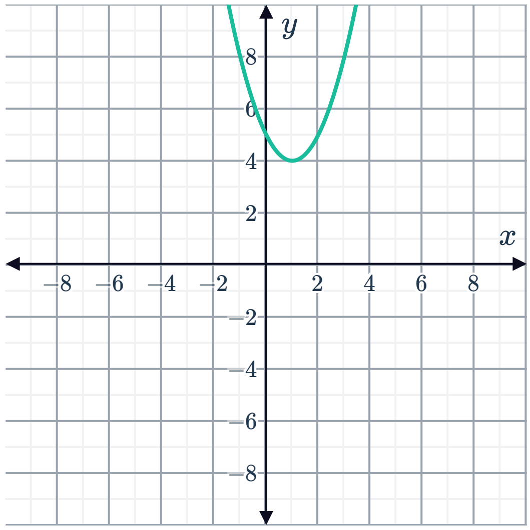
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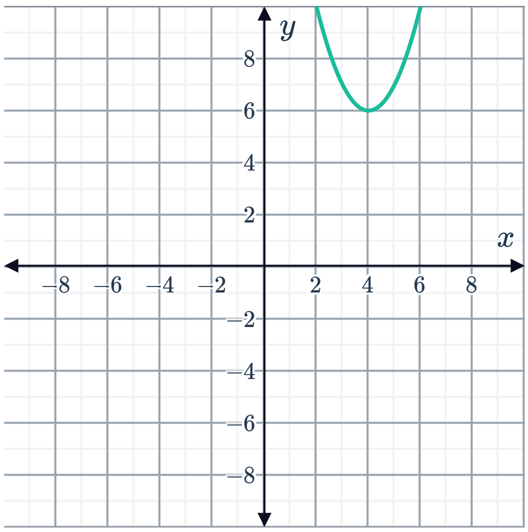
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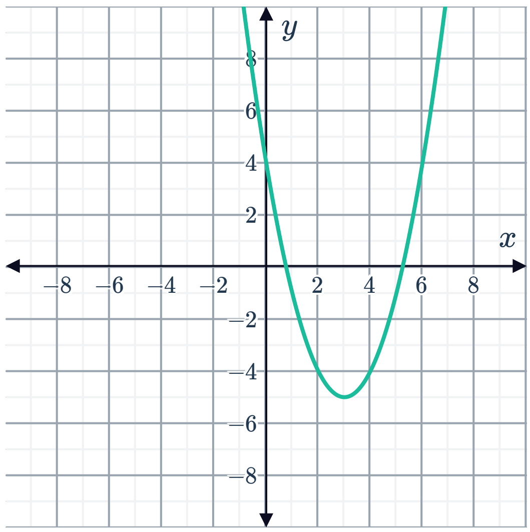
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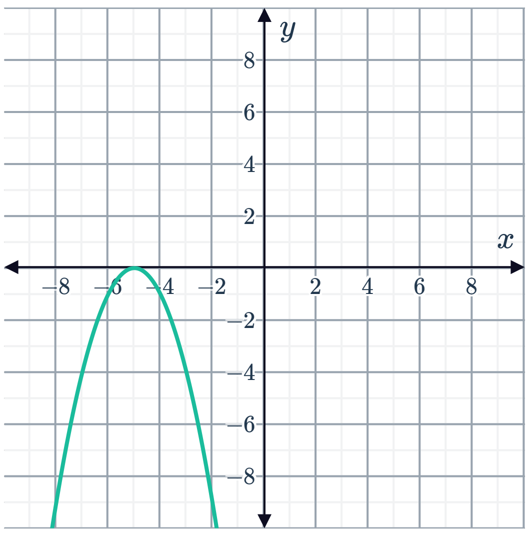
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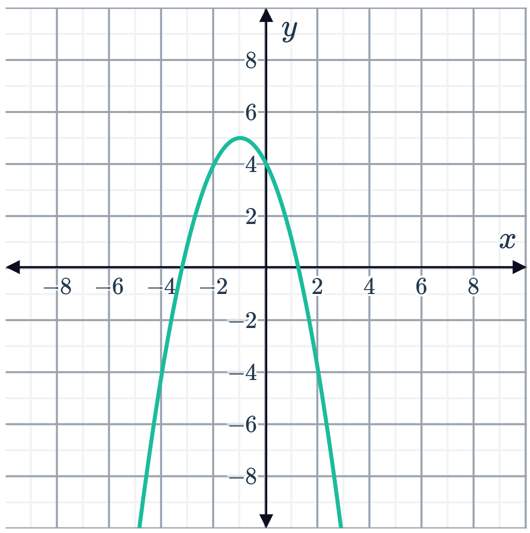
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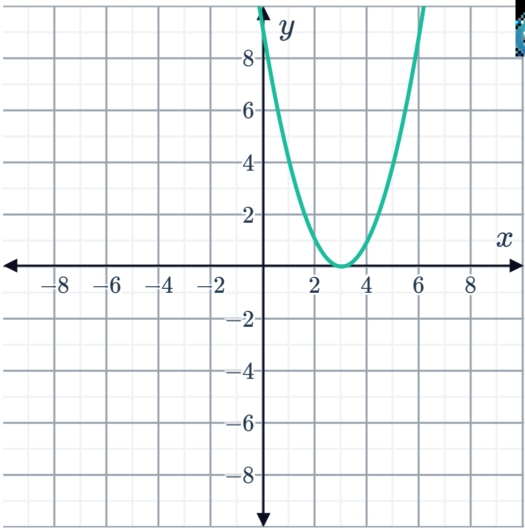
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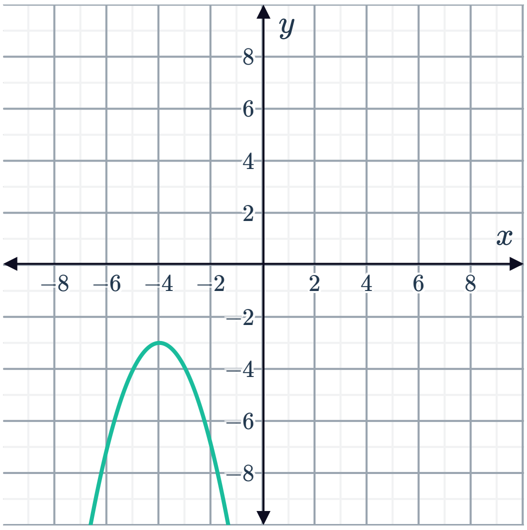
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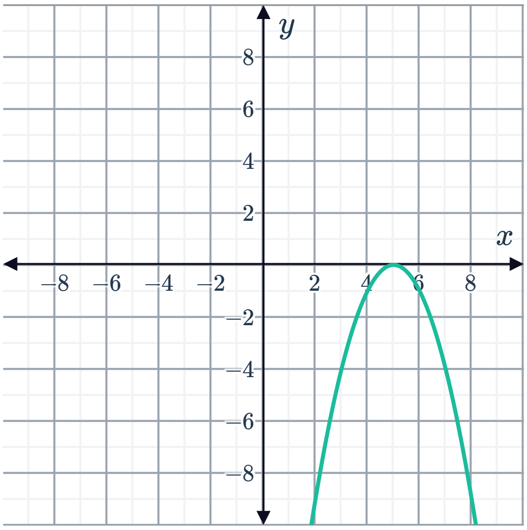
k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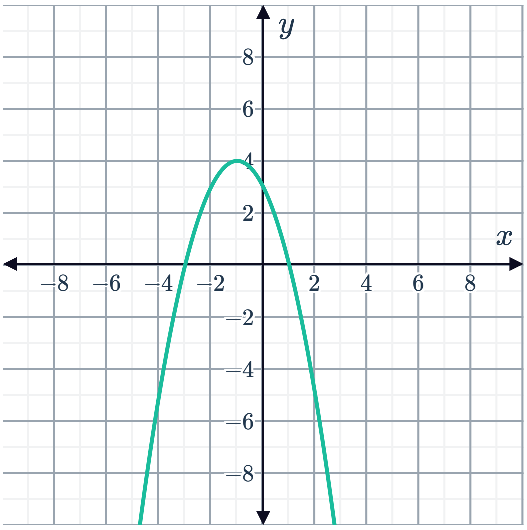
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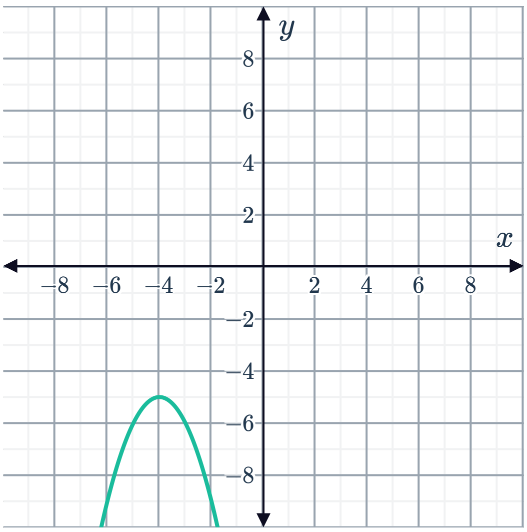
m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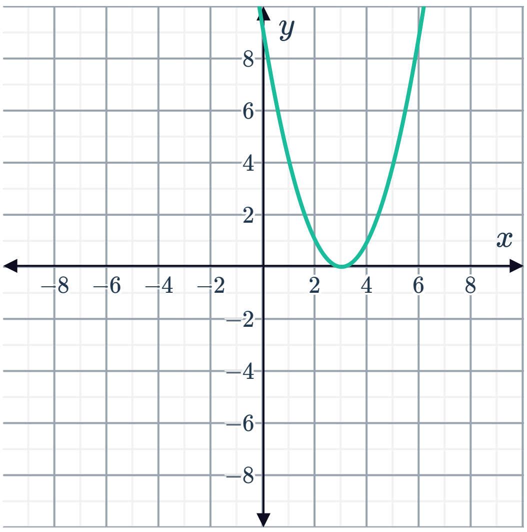
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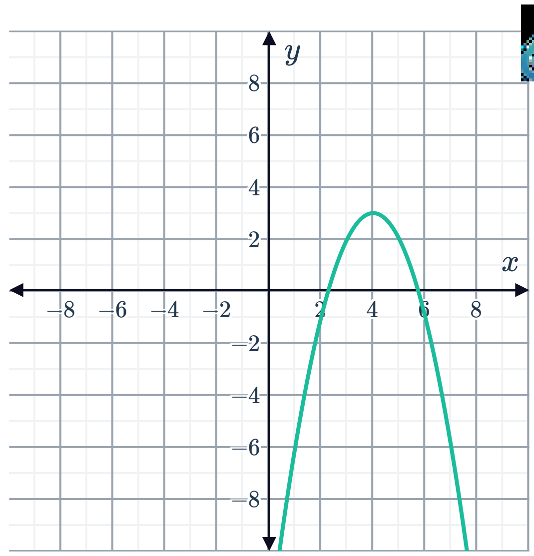
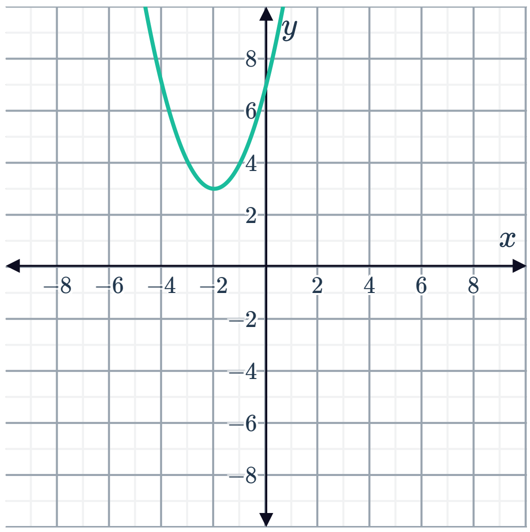


p



22 Consider the graph of the quadratic function:

$$y = m - 9x - 3x^2$$

a Find the possible values of m, if the graph has no x-intercepts.

b State the largest possible integer value of m.

23 Determine the value(s) of k for which the graph of $y = 4x^2 - 4x + k - 15$ just touches the x-axis.

Application

24 Consider a right-angled triangle with side lengths x units, $x + p$ units and $x + q$ units, ordered from shortest to longest. No two sides of this triangle have the same length.

a Complete the statement:
p and q have lengths such that
$0 < \square < \square$.

b Write a quadratic equation that describes the relationship between the sides of the triangle in terms of x.

c Find the discriminant of this quadratic equation.

d State the number of real solutions to the quadratic equation. Explain your answer.

e Find the value of x in terms of p when $q = 2p$.